

Function	Dim	Index	GA	ASTER-GA	FA	ASTER-FA	DE	ASTER-DE	jDE	L-SHADE
F1	10	Ave	8.39e+03	9.03e+00	5.91e+03	2.72e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00
		Std	2.72e+03	2.10e+00	1.72e+03	9.31e+02	0.00e+00	0.00e+00	0.00e+00	0.00e+00
	20	Ave	4.01e+04	5.96e+01	3.13e+04	1.19e+04	0.00e+00	0.00e+00	0.00e+00	0.00e+00
		Std	8.89e+03	9.17e+00	3.03e+03	2.55e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F2	10	Ave	1.61e+03	9.35e+00	2.00e+02	8.01e+01	5.83e+00	6.78e+00	5.33e+00	7.14e+00
		Std	7.06e+02	1.26e+01	6.22e+01	3.03e+01	2.67e+00	2.48e+00	2.49e+00	2.65e+00
	20	Ave	3.36e+03	5.80e+01	1.81e+03	3.90e+02	4.66e+01	4.66e+01	4.91e+01	4.88e+01
		Std	1.04e+03	1.14e+01	4.54e+02	1.14e+02	2.09e+00	8.87e+00	0.00e+00	1.06e+00
F3	10	Ave	5.37e-01	3.79e-04	2.01e-01	7.31e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00
		Std	1.75e-01	8.58e-05	3.72e-02	1.86e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00
	20	Ave	1.97e+00	2.27e-03	1.17e+00	3.01e-01	0.00e+00	0.00e+00	0.00e+00	0.00e+00
		Std	2.44e-01	3.59e-04	1.90e-01	6.14e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F4	10	Ave	6.47e+01	1.72e+01	9.20e+01	7.56e+01	3.17e+01	3.03e+01	2.03e+01	9.80e+00
		Std	1.82e+01	3.54e+00	1.66e+01	1.50e+01	3.67e+00	6.82e+00	4.55e+00	2.80e+00
	20	Ave	3.06e+02	7.41e+01	3.47e+02	3.06e+02	1.57e+02	1.56e+02	7.19e+01	3.27e+01
		Std	3.91e+01	1.87e+01	3.52e+01	2.50e+01	1.96e+01	1.92e+01	1.25e+01	5.97e+00
F5	10	Ave	2.92e+00	9.52e-03	2.46e+00	1.22e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
		Std	6.60e-01	2.45e-03	5.37e-01	3.87e-01	0.00e+00	0.00e+00	0.00e+00	0.00e+00
	20	Ave	1.16e+01	6.63e-01	1.32e+01	6.59e+00	0.00e+00	2.98e-03	0.00e+00	0.00e+00
		Std	3.49e+00	1.21e+00	1.65e+00	1.33e+00	0.00e+00	1.63e-02	0.00e+00	0.00e+00
F6	10	Ave	4.26e+08	5.62e+04	8.20e+06	4.77e+06	2.83e-01	3.39e-01	3.95e-01	2.99e-01
		Std	2.76e+08	1.84e+04	4.75e+06	3.44e+06	1.70e-01	2.63e-01	2.21e-01	1.39e-01
	20	Ave	6.47e+09	1.67e+06	1.09e+09	3.53e+08	4.38e+03	1.31e+04	9.10e+03	9.26e+02
		Std	1.68e+09	4.21e+05	3.13e+08	1.19e+08	3.46e+03	9.22e+03	6.43e+03	8.23e+02
F7	10	Ave	8.74e+02	3.36e+01	1.88e+02	1.63e+02	2.22e+01	2.21e+01	4.33e+00	2.22e-01
		Std	4.07e+02	3.05e+00	5.68e+01	4.48e+01	6.14e+00	7.31e+00	2.03e+00	1.83e-01
	20	Ave	6.52e+03	5.57e+01	1.79e+03	9.74e+02	4.07e+01	3.66e+01	2.88e+01	2.13e+01
		Std	2.24e+03	1.36e+01	4.38e+02	2.50e+02	3.93e+00	3.84e+00	2.24e+00	3.29e-01
F8	10	Ave	3.76e+10	3.00e+01	7.69e+02	6.65e+02	1.71e+00	8.46e+00	5.70e+00	3.92e+00
		Std	1.16e+11	4.04e+00	4.77e+02	4.64e+02	3.72e-01	7.61e+00	2.29e+00	7.55e+00
	20	Ave	5.18e+13	1.30e+02	1.31e+07	2.23e+06	2.63e+01	2.68e+01	2.36e+01	2.05e+01
		Std	1.20e+14	9.45e+01	2.34e+07	3.92e+06	1.73e+00	3.18e+00	5.15e+00	3.81e+00
F9	10	Ave	1.25e+03	1.94e+02	4.93e+02	3.95e+02	1.20e+02	8.38e+01	9.58e+01	1.20e+01
		Std	3.97e+02	1.80e+02	4.90e+01	3.36e+01	1.72e+02	1.54e+02	1.62e+02	6.56e+01
	20	Ave	6.04e+03	3.40e+02	1.22e+03	5.32e+02	3.36e+02	3.36e+02	3.36e+02	3.36e+02
		Std	1.63e+03	1.80e+00	1.82e+02	7.56e+01	1.21e-13	2.03e-13	1.19e-13	2.28e-13
F10	10	Ave	6.04e+02	2.43e+02	2.79e+02	2.48e+02	2.21e+02	2.21e+02	2.21e+02	2.21e+02
		Std	1.85e+02	4.29e+01	1.20e+01	4.54e+00	4.55e-02	6.42e-01	1.07e-01	3.01e-01
	20	Ave	4.53e+03	1.16e+03	7.45e+02	5.13e+02	3.99e+02	4.12e+02	4.02e+02	4.00e+02
		Std	8.18e+02	9.42e+02	5.89e+01	1.79e+01	1.21e+01	2.77e+01	1.82e+01	1.39e+01
F11	10	Ave	6.67e+02	3.06e+01	1.09e+02	4.08e+01	1.52e-14	1.52e-14	0.00e+00	1.52e-14
		Std	3.08e+02	1.60e+02	2.92e+01	6.13e+00	8.30e-14	8.30e-14	0.00e+00	8.30e-14
	20	Ave	1.16e+04	2.56e+00	1.64e+03	1.66e+02	4.55e-13	4.45e-01	4.40e-13	4.55e-13
		Std	5.27e+03	2.95e-01	5.61e+02	2.97e+02	0.00e+00	2.44e+00	8.30e-14	0.00e+00
F12	10	Ave	5.52e+02	1.66e+02	1.72e+02	1.72e+02	1.65e+02	1.65e+02	1.65e+02	1.65e+02
		Std	1.11e+02	1.10e+00	1.06e+01	1.07e+00	6.55e-01	8.23e-01	6.76e-01	7.97e-01
	20	Ave	1.49e+03	2.71e+02	5.43e+02	3.11e+02	2.40e+02	2.48e+02	2.39e+02	2.39e+02
		Std	2.48e+02	1.78e+01	5.03e+01	2.70e+01	5.36e+00	9.74e+00	1.43e+00	1.59e+00

Table 1: Evaluation results on the CEC 2022 test suite functions (F1–F12) at $D = 10$ and $D = 20$. Bold indicates the best (lowest) average value in each row; ties are all bolded. Budget: 100,000 FEs ($D = 10$) and 200,000 FEs ($D = 20$), 30 independent runs.